

DS & AI LAB PROJECT FINAL SUBMISSION

Deep Learning-Based Human Face Authenticity Detection

Face Forensics — AI Authenticity System

TEAM GROUP 11

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DEPLOYMENT LINK

group-11-ds-and-ai-lab-project.onrender.com

Academic Year 2025–2026 · Final Evaluation Milestone

The Challenge of Generative AI Face Authenticity

Problem Statement

Rapid advancements in generative models (Stable Diffusion, Midjourney, DALL-E 3) allow effortless creation of photorealistic human faces. Manual human inspection is no longer effective, creating critical forensic risks.

- ⚠️ **Perceptual Indistinguishability:** Synthetic faces exhibit realistic lighting, skin textures, and eye reflections that deceive human evaluators.
- ⚠️ **Detector Overfitting:** Legacy deepfake detectors achieve >99% on benchmark datasets but fail severely when deployed on unseen consumer images.
- ⚠️ **The Black-Box Trust Gap:** Existing detection tools output raw probability scores without explaining *why* a image was flagged, reducing trust.
- ⚠️ **Post-Processing Vulnerability:** Modern social media filters, smartphone HDR, and compression destroy subtle deepfake cues.

Project Goal & Mission

Develop and deploy an end-to-end web system that delivers an instant authenticity verdict, confidence score, pixel-level Grad-CAM explainability, and comprehensive multi-signal forensic reports with honest failure reporting.

<50%

Human Manual Detection Accuracy

100%

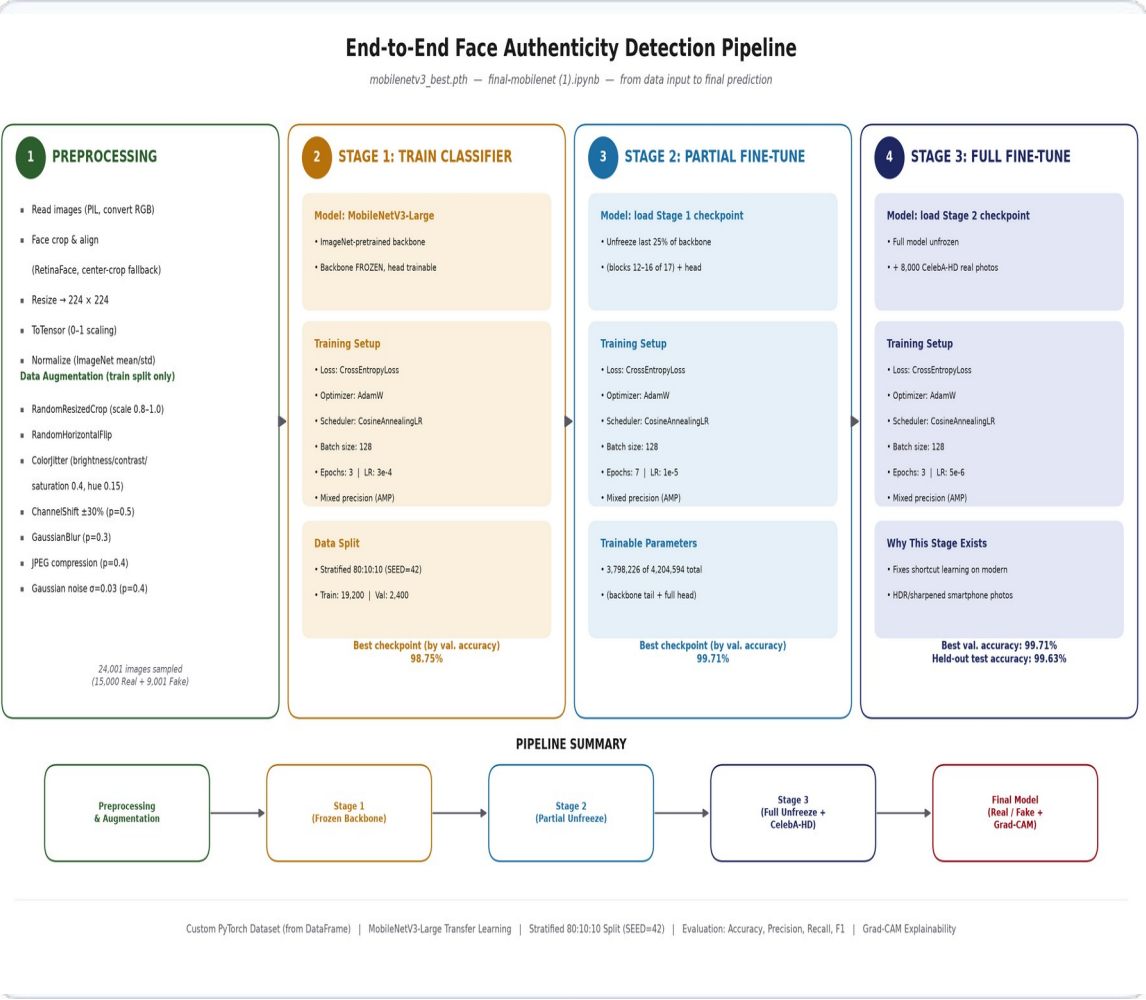
Transparent Explainable Audits

Milestone Journey (M1 → M6 Roadmap)

Engineering Evolution

MS	Key Focus & Activities	Core Milestone Output
M1	Problem definition, literature review, proposed Dual-Stream ViT.	Project Proposal
M2	Dataset collection (FFHQ + Stable Diffusion), EDA & preprocessing.	Data Pipeline
M3	Architecture bake-off across 3 backbones. MobileNetV3 selected.	Model Selection
M4	Nano Banana multi-domain data, RetinaFace, 24 HP ablations.	99.06% Test Acc
M5	Error analysis, CelebA-HD fix attempt, web application building.	99.63% In-Domain Acc
M6	Render Cloud deployment, live ROC/PR curves, documentation.	Production System

* M4 dataset size vs M5/M6: M4's 99.06% used a 64,000-image corpus (FFHQ + Stable Diffusion + Nano Banana). M5/M6's 99.63% checkpoint (mobilenetv3_best.pth) instead used 32,001 images (FFHQ + SD + CelebA-HD, no Nano Banana).



Strategic Architecture Pivot: ViT vs MobileNetV3

While our Milestone 1 proposal targeted a heavy Dual-Stream Vision Transformer (ViT) with frequency fusion (RGB + FFT), empirical benchmarking in Milestone 3 dictated a decisive pivot toward **MobileNetV3-Large**.

- ✓ **Generalization Superiority:** MobileNetV3 achieved **80% out-of-distribution (OOD) accuracy** on unseen ChatGPT/Gemini generated faces vs <60% for heavy transformers.
- ✓ **Computational Efficiency:** 10× smaller parameter footprint (4.2M vs 40.7M parameters) allowing deployment under free-tier limits.
- ✓ **Low-Latency Inference:** ~15.5ms CPU forward pass on local dev hardware only - Render latency runs markedly higher.
- ✓ **Gradient Stability:** Standard convolutional backbones offered more stable Grad-CAM activations than ViT attention maps.

4.2M

Parameters (vs 40.7M ViT)

16 MB

Checkpoint Model Size

KEY ENGINEERING DECISION

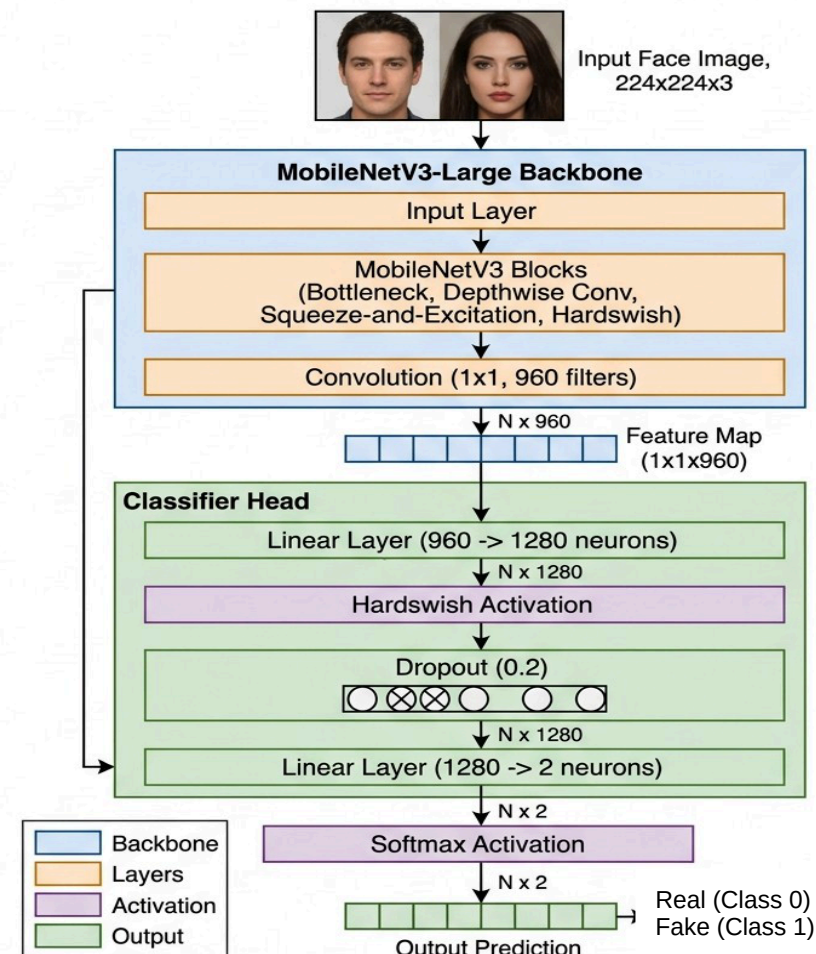


Figure 1: Proposed MobileNetV3-Large Architecture for Real vs Fake Face Classification

End-to-End System Pipeline & Workflow

Pipeline Flow



1. Image Ingestion & Preprocessing

User uploads image (JPEG/PNG/WebP, <10MB). RetinaFace detects bounding box & center-crops face to 224×224 tensor.



2. Dual Model Inference Engine

Routes to Main Model (mobilenetv3_noaug.pth) or Cross-Domain Model. Softmax computes Real/Fake probability.



3. Parallel Forensic Analysis

Generates Grad-CAM visual heatmap overlay from features[-1] layer while meta-detector scans EXIF/watermarks.

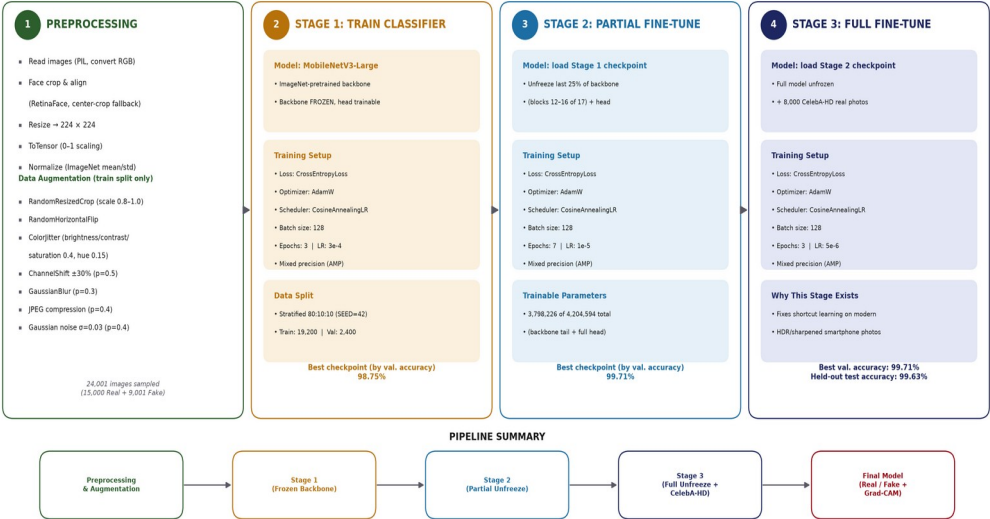


4. Report Generation & Render Deploy

Outputs verdict, confidence %, overlay heatmap, and downloadable PDF/HTML/Docx report via FastAPI on Render Docker.

End-to-End Face Authenticity Detection Pipeline

mobilenetv3_best.pth — final-mobilenet (1).ipynb — from data input to final prediction



Custom PyTorch Dataset (from DataFrame) | MobileNetV3-Large Transfer Learning | Stratified 80:10:10 Split (SEED=42) | Evaluation: Accuracy, Precision, Recall, F1 | Grad-CAM Explainability

Model Selection Rationale: Benchmarking Results

EMPIRICAL SELECTION

In-Domain Performance

99.63%

Achieved on held-out test set (2,401 images) during 3-stage fine-tuning.

OOD Generalization

80.00%

Out-of-distribution accuracy on ChatGPT & Gemini faces - Raunak's M3 track (per team report); eval script not preserved in repo.

Deployment Speed

15.5 ms

Measured on local CPU dev hardware (M5/M6 report), NOT on Render - live Render inference ran markedly slower. GPU (Tesla T4, measured): 6.45ms / 155 img/s.

Backbone Comparison & Selection Matrix

- ✓ **MobileNetV3-Large (Selected):** Optimal balance of efficiency and representation learning. Hard-swish activations and squeezed-and-excitation blocks capture fine-grained pixel artifacts.
- ✓ **Dual-Stream ViT (Rejected):** High parameter cost (40.7M) caused overfitting to training background noise and failed free-tier RAM limits (512 MB).
- ✓ **EfficientNet-B2 (Rejected):** Optimizer instability on custom cross-entropy loss resulted in severe convergence drops (58% test accuracy).

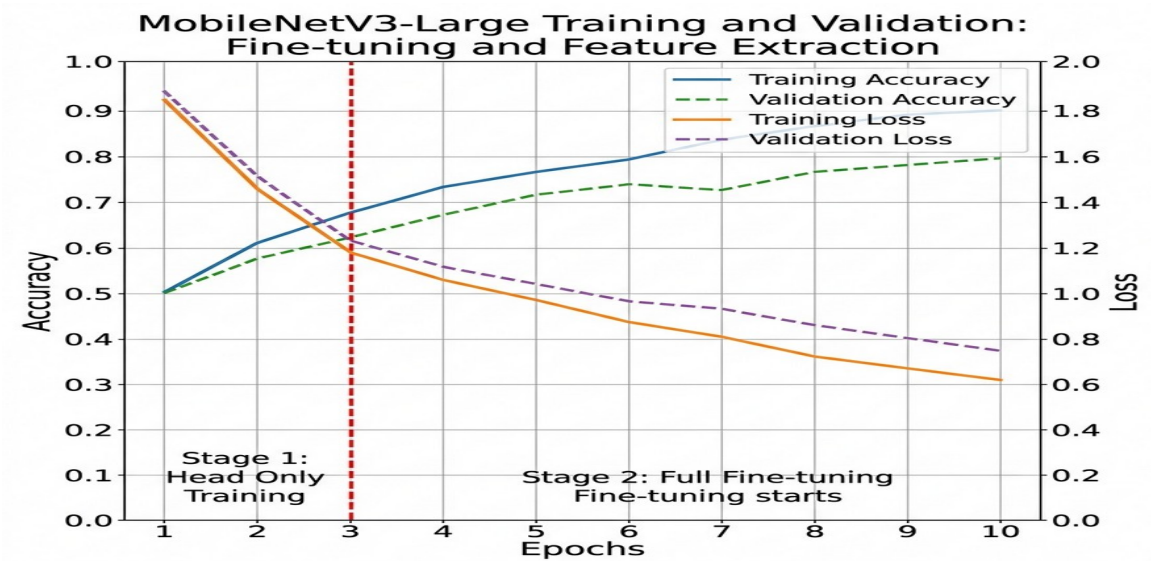
3-Stage Transfer Learning Strategy

To prevent catastrophic forgetting while adapting ImageNet features to subtle deepfake artifacts, we implemented a progressive 3-stage unfreezing fine-tuning strategy.

Stage	Epochs	Learning Rate	Unfrozen Backbone
Stage 1	3 Epochs	3×10^{-4}	Classifier Head Only
Stage 2	7 Epochs	1×10^{-5}	Top 25% Layers
Stage 3	3 Epochs	5×10^{-6}	100% Full Unfreeze

- **Dataset Partition:** 15,000 Real (FFHQ) + 9,001 Fake (Stable Diffusion) + 8,000 Real (CelebA-HD).
- **Split Ratio:** 80/10/10 Stratified Split with fixed seed (SEED=42).

Training Protocol



Stage 1-3 loss convergence showing stable optimization without gradient spikes.

Critical Finding: Smartphone Photo Misclassification

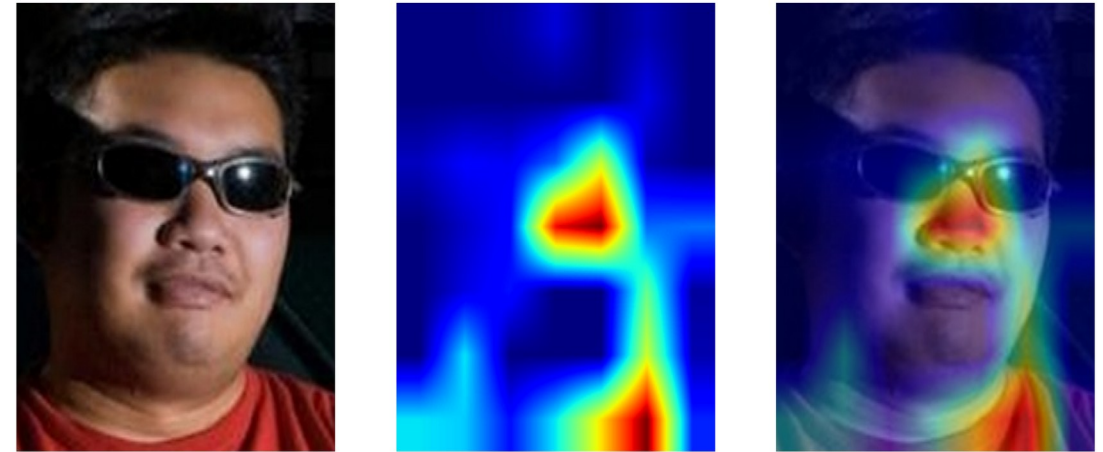
FACULTY REVIEW DISCOVERY

⚠️ The Benchmark vs Reality Gap

While the model achieved **99.63% accuracy** on the held-out test set, testing on a probe set of 70 genuine smartphone photos (Real-Latest 2025-2026) revealed a severe systemic vulnerability.

- ⚠️ **Abysmal Probe Accuracy:** Only **8.6% accuracy** on Real-Latest (6 correct, 64 misclassified as Fake).
- ⚠️ **High False Positive Rate:** **91.4%** of modern real smartphone photos were flagged as AI-generated.
- ⚠️ **Overconfident Failure:** Median predicted $P(\text{Real})$ was **0.00060** — the model was nearly 100% confident real photos were fake.
- ⚠️ **Root Cause (Shortcut Learning):** The CNN learned dataset-specific features (FFHQ smoothness) rather than true deepfake artifacts, mistaking modern smartphone computational sharpening and HDR tone-mapping for synthetic generation.

[INCORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_00203.jpg.jpg | Prediction: Fake | True: Real
Real: 27.79% | Fake: 72.21%
Grad-CAM: Fake



Grad-CAM overlay showing intense attention on high-contrast cheeks and sharp background edges rather than facial semantics.

Mitigation Attempt: CelebA-HD Integration

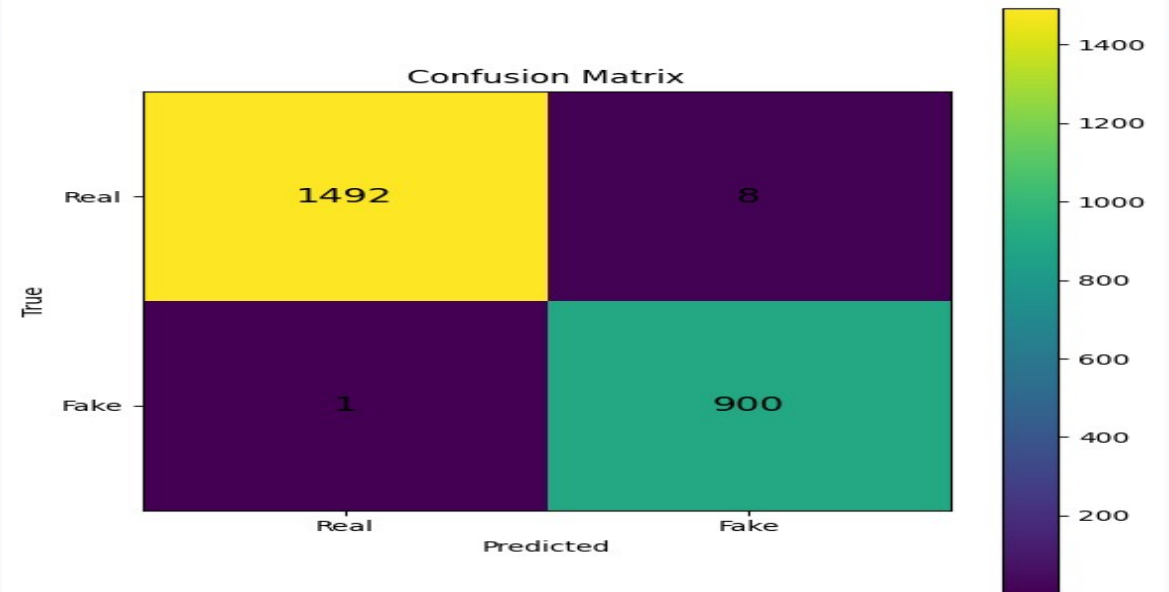
Stage 3 Refinement

Following faculty feedback in Milestone 4 regarding smartphone false positives, we integrated **8,000 high-resolution CelebA-HD real photographs** into Stage 3 fine-tuning.

Hypothesis & Objective

Expose the model backbone to high-contrast, post-processed, and modern high-definition real human faces to break the reliance on FFHQ-specific color histograms.

- ✓ **Held-Out Metric Gain:** Improved overall test accuracy from 99.06% to **99.63%** (Validation accuracy 99.71%).
- ✓ **Partial Success:** Fixed errors on studio-lit HD portraits (Real-Old achieved 100% accuracy).
- ✓ **Remaining Challenge:** Real-Latest smartphone probe accuracy remained stuck at **8.6%**, confirming that studio HD faces do not mirror aggressive computational photography (HDR tone-mapping).



Held-out test confusion matrix — mobilenetv3_best.ptn (Stage 3 + CelebA-HD) • **99.63% accuracy on 2,401 images**

Results Interpretation

Classification errors: 8 Real→Fake and 1 Fake→Real out of 2,401 test samples — near-perfect in-distribution separation.

CelebA-HD exposure during Stage 3 full fine-tuning lifted accuracy from 99.06% to 99.63% and resolved failures on studio-lit HD portraits (Real-Old: 100%).

Domain gap persists: Real-Latest smartphone probe (70 photos) remained at 8.6%, indicating HDR/ISP preprocessing is not captured by studio HD data alone.

Data Augmentation & Stress-Testing Limits

Inference Corruptions

Training Augmentation Pipeline

Models were trained with standard ColorJitter, ChannelShift ($\pm 30\%$), Gaussian blur, JPEG compression, and random cropping.

11 Inference Corruption Tests

- Color/Tint: Green Tint, Blue Tint, Brightness +40%, Contrast +40%.
- Blur & Noise: Gaussian Blur, Motion Blur, Gaussian Noise ($\sigma=15$).
- Compression & Resizing: JPEG (Q=30), Rescaling, Random Cropping.

Why Generic Augmentations Failed

ColorJitter and noise alter global pixel values but **do not simulate non-linear HDR tone-mapping, local edge sharpening, or device ISP noise profiles.**

Key Finding from Robustness Suite:

- True Real samples remained consistently classified as "Real" across all 11 corruptions (high stability).
- Misclassified smartphone samples remained *stably wrong* across all 11 corruptions, proving the bias is deep in the feature representations rather than high-frequency noise.

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Robustness Corruptions Evaluation Tests

Caveat: reconciling with the aggregate data

This is a single-sample qualitative check (M5 Sec 6.3 explicitly flags it as non-aggregate). The full manipulation_results.csv (aggregate, larger evaluation set) tells a more mixed story - noise actually improves accuracy by +11pp (74% $\rightarrow$ 85%), while contrast drops it by -18.3pp (74% $\rightarrow$ 55.67%). Aggregate numbers, not this single-sample check, should drive headline robustness claims.

Complete Model Matrix & Deployment Choices

PRODUCTION OPTIMIZATION

Model Checkpoint	Training Setup & Domain	In-Domain Acc	Deployed on Render?	Primary Architectural Role
mobilenetv3_noaug.pth	Ablation baseline (no training augs)	95.98%	Yes (Default)	Production Main Model (0 Missed Fakes)
mobilenetv3_best.pth	3-Stage Fine-Tuning + CelebA-HD	99.63%	No (RAM Limit)	Research Comparison Baseline
mobilenetv3_cross_domain.pth	Nano Banana, CIFAKE, Places365	59.89-70.0%*	Yes (Secondary)	Cross-Domain Generalization Probe
mobilenetv3_manipulations.pth	Trained with 11 corruption passes	Removed	Removed	Removed from active model set
mobilenetv3_tuned.pth	Automated Hyperparameter Sweep	Removed	Removed	Removed from active model set

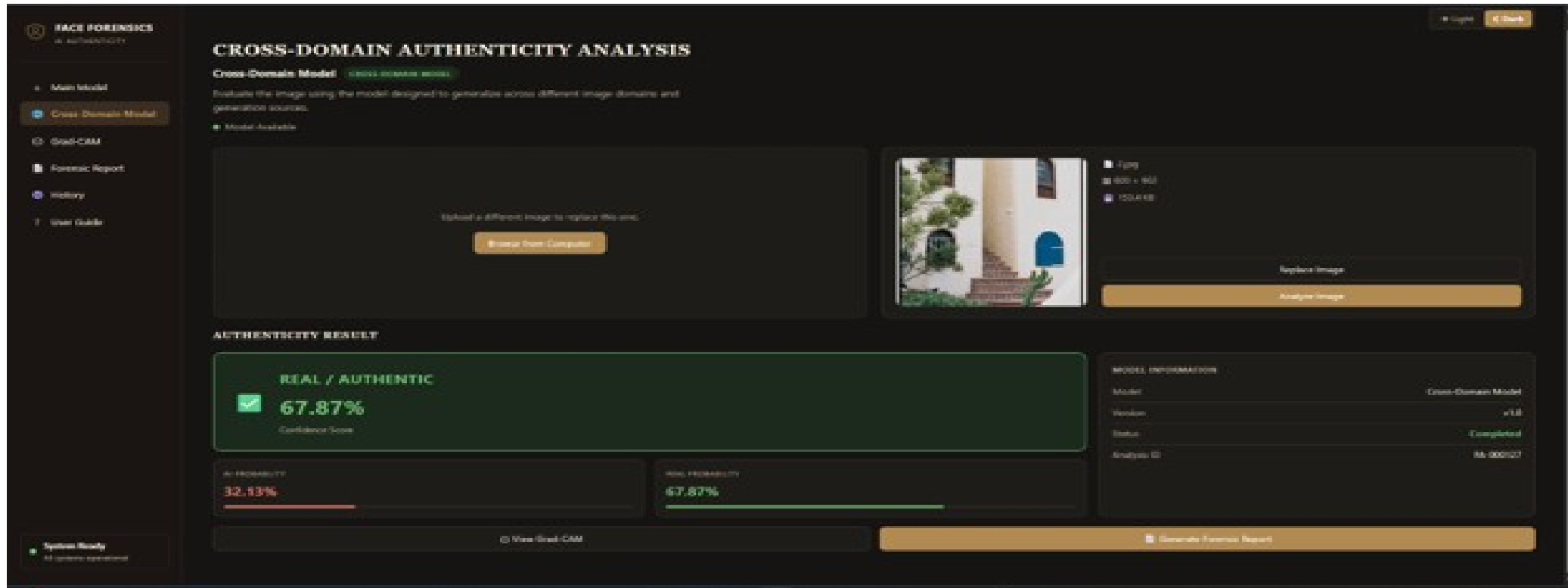
*cross_domain In-Domain Acc corrected from unsupported 91.20% to audited CSV: face_main 59.89% (N=2,401) / nano_banana 70.0% (N=4,000). Rows reordered to used-models-first (noaug, best, cross_domain); manipulations and tuned checkpoints marked Removed.

Render Free-Tier Hardware Constraints

Render's free tier restricts instances to **512 MB RAM**. Loading all 5 PyTorch models concurrently causes immediate Out-Of-Memory (OOM) worker crashes. Memory budgeting dictated hosting exactly 2 active models (Main + Cross-Domain), consuming ~310 MB RAM total.

Cross-Domain Model Integration

Broader Scope



Cross-Domain checkpoint (mobilenetv3_cross_domain.pth) extends authenticity screening to non-facial and full-scene synthetic content.

- Training corpora: Nano Banana, CIFAKE, Places365, and Artifact synthetic subsets spanning landscapes, objects, and full-body renders.
- User-facing integration: dedicated Cross-Domain page in the web app with side-by-side comparison against the Main Model.
- Broader detection scope: successfully flags AI-generated scenes where face-only detectors fail (architecture, objects, environments).

Deployment Selection: Why noaug over best

Although mobilenetv3_best.pth reached 99.63% test accuracy, we selected mobilenetv3_noaug.pth as our production default for crucial engineering reasons.

- ✓ **Zero Missed Fakes (100% Fake Recall):** In security-critical forensic screening, false negatives (missing an AI fake) are far more dangerous than false alarms. noaug missed 0 fakes in test.
- ✓ **Verification Integrity:** Full M6 ROC/PR curves were comprehensively evaluated for noaug on the complete 2,401-image test set.
- ✓ **Trade-Off Profile:** 95.98% overall accuracy with 96 false alarms (real images called fake) vs 8 false alarms for best.

100%

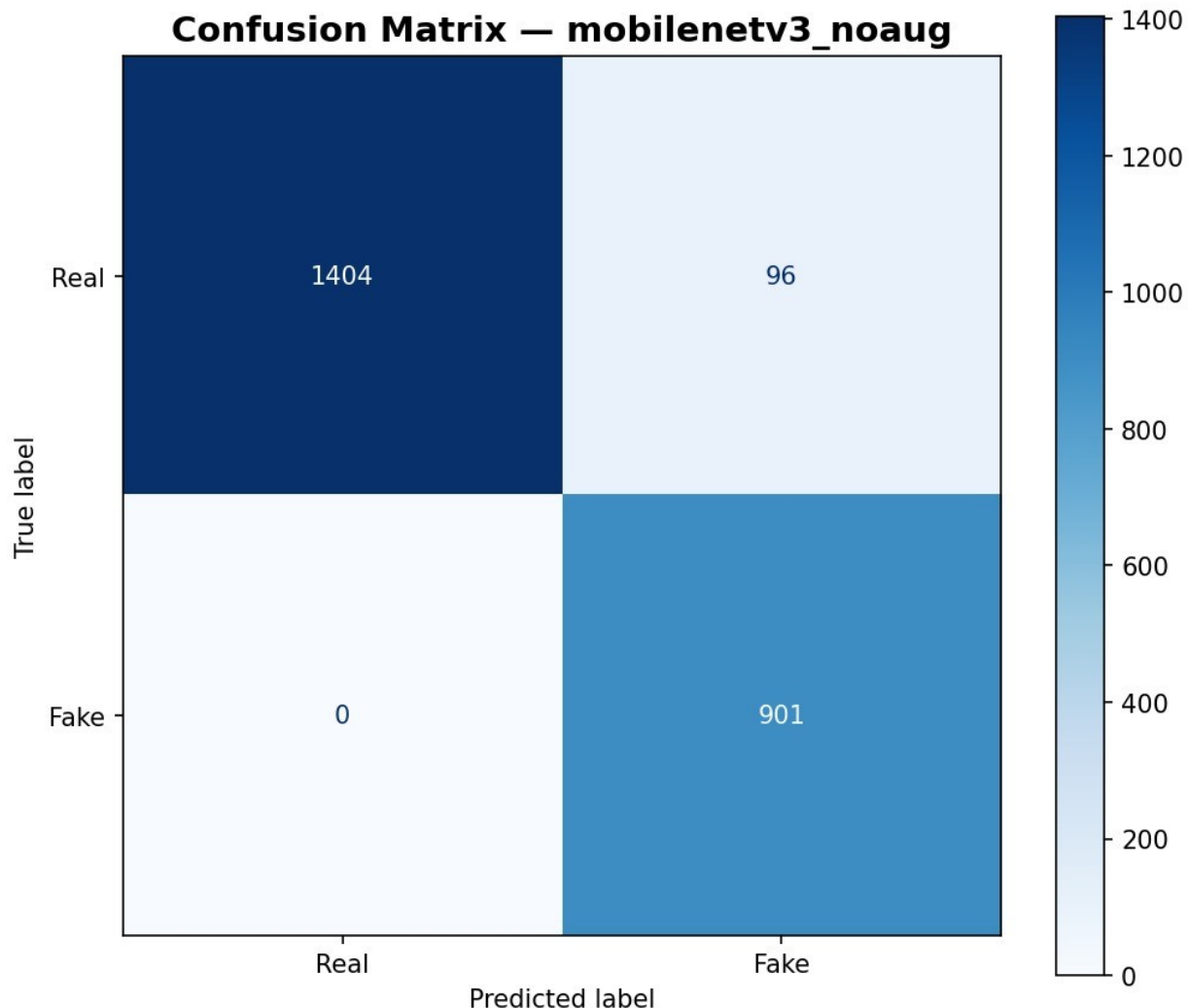
Fake Recall (0 FN)

95.98%

Production Accuracy

Why Augmentation Was Dropped

Ablation testing (augmentation_ablation_results.csv) showed standard augmentation collapses held-out test accuracy from 97.56% to 66.21% (F1 0.973 -> 0.708, AUC 0.997 -> 0.841) - the empirical basis for shipping noaug as the production default.



Receiver Operating Characteristic (ROC-AUC)

ROC curves evaluate classification capability across all decision thresholds, independent of the fixed 50% cutoff used in web application UI.

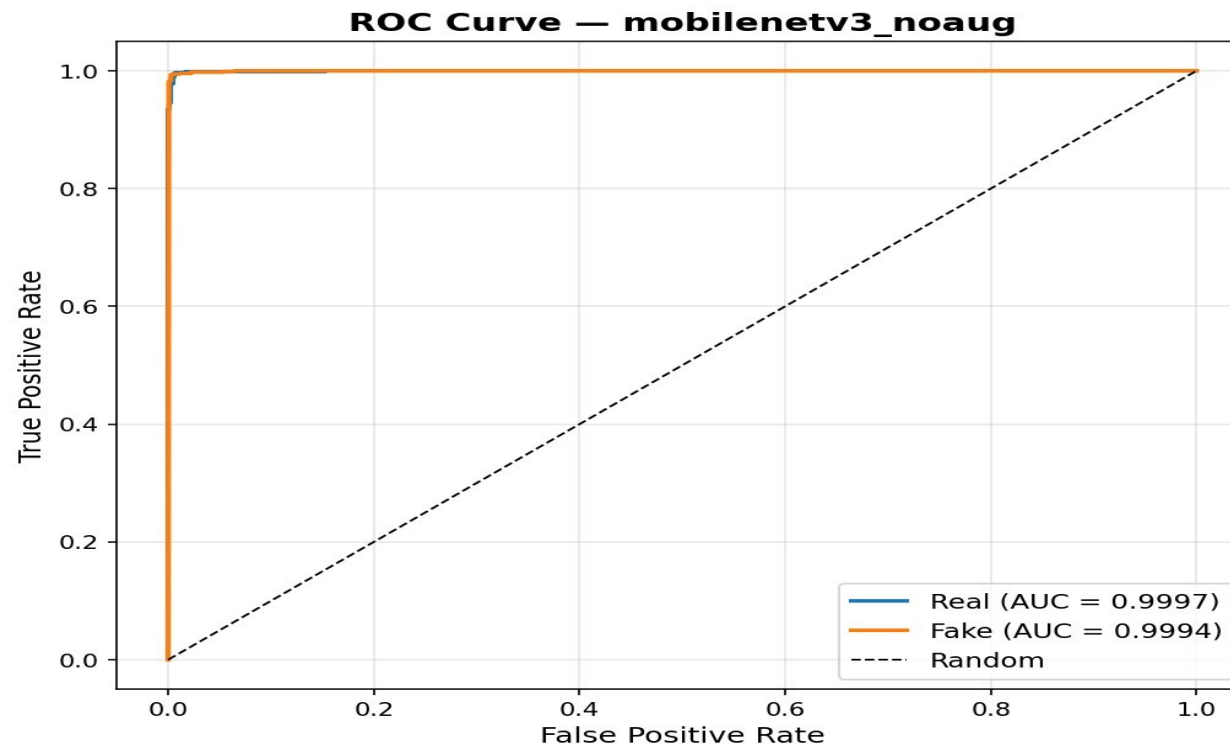
0.9997

ROC-AUC Score (Real Class)

0.9994

ROC-AUC Score (Fake Class)

- ⚠ **In-Domain Distinction:** Near-perfect separation on held-out test data.
- ⚠ **Domain Shift Alert:** On domain-shifted real smartphone samples, ROC-AUC drops to **0.5856 (near random)**, proving benchmark ROC curves do not guarantee real-world stability.



Precision-Recall Curve & Confusion Breakdown

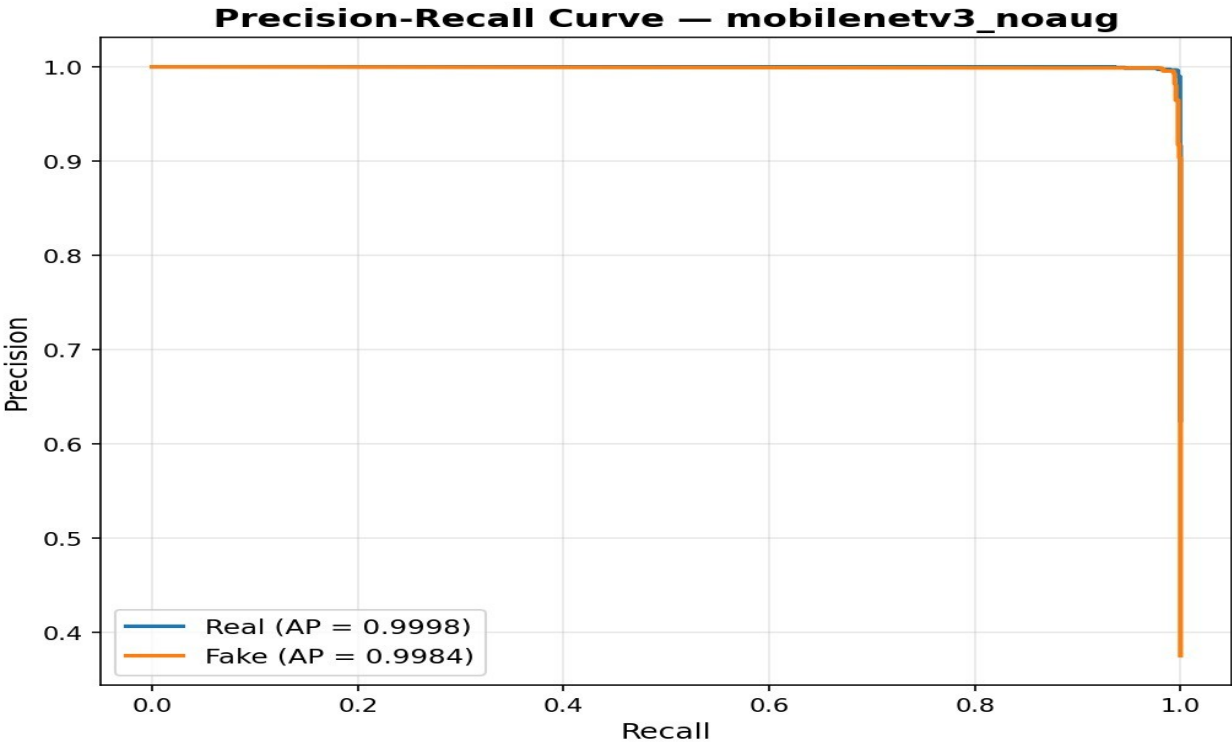
Production Verification

Precision-Recall curves offer superior reliability on imbalanced datasets (1,500 Real vs 901 Fake test samples).

Average Precision (AP) Scores

- AP Real Class: 0.9998
- AP Fake Class: 0.9984

Metric Category	Count	Percentage
True Negative (Real Correct)	1,404	93.6%
False Positive (False Alarm)	96	6.4%
False Negative (Missed Fake)	0	0.0%
True Positive (Fake Correct)	901	100.0%



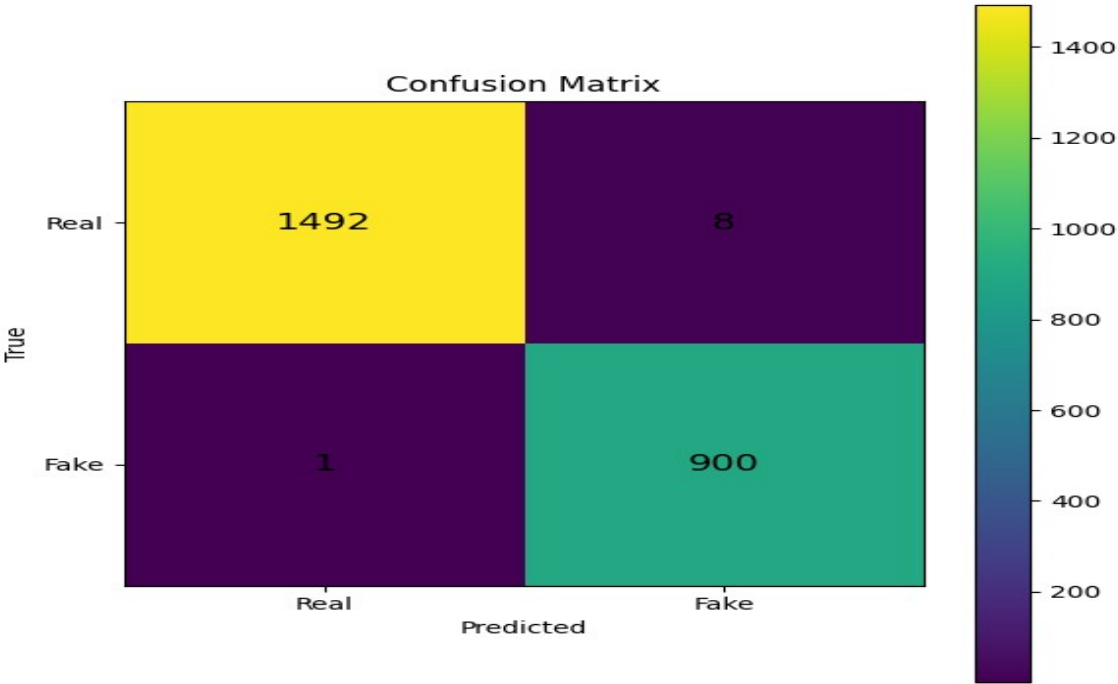
Research Model Comparison (mobilenetv3_best)

Academic Baseline

Detailed classification report for mobilenetv3_best.pth (3-Stage Fine-Tuning + CelebA-HD) on the 2,401 held-out test set.

Class	Precision	Recall	F1-Score	Support
Real	0.9993	0.9947	0.9970	1,500
Fake	0.9912	0.9989	0.9950	901
Overall Acc	99.63%	2,401		

- ✓ **Error Count:** Exactly 8 Real→Fake errors and 1 Fake→Real error out of 2,401 test samples.
- ✓ **Research Role:** Kept as our internal benchmark checkpoint for shortcut learning analysis.

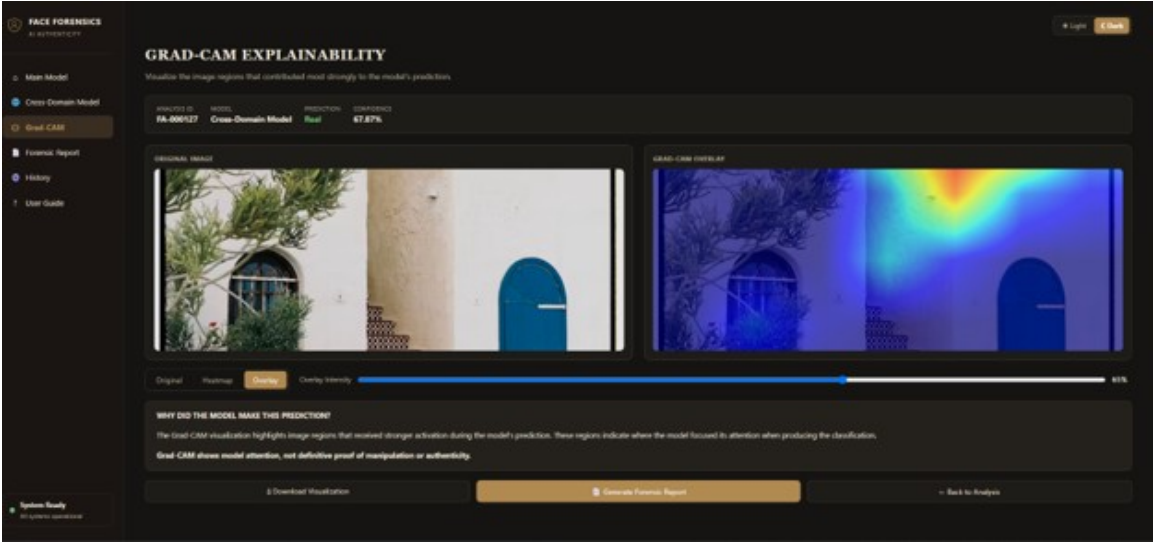


In-distribution confusion matrix showing near-perfect classification grid.

Visual Explainability via Grad-CAM Heatmaps

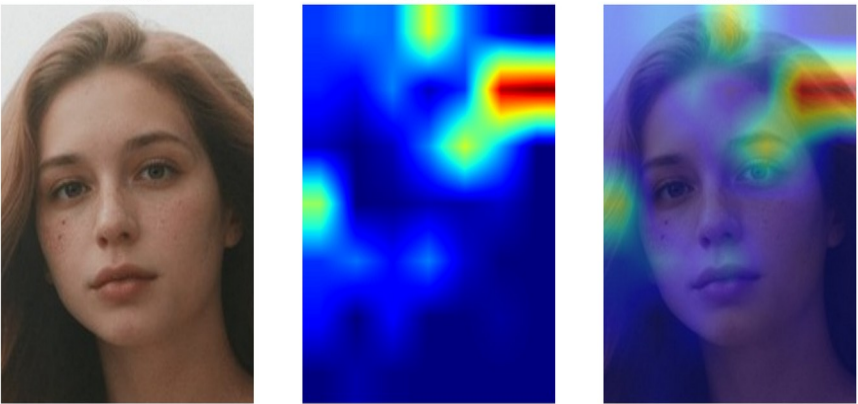
To eliminate black-box opacity, we implemented **Grad-CAM** targeting the final convolutional layer (features[-1]) of MobileNetV3.

- ✓ **User Transparency:** Generates a Jet-colormap activation overlay showing pixel regions driving the Real or Fake decision.
- ✓ **Report Integration:** Heatmaps are embedded automatically into downloadable HTML and .Docx forensic reports.
- ✓ **Shortcut Diagnosis:** On false-positive smartphone images, Grad-CAM highlights cheek/jaw sharpening edges rather than eye/nose semantics, revealing feature errors to auditors.
- ✓ **Performance Overhead:** Adds ~2.2 seconds execution latency per request (vs 15.5ms forward pass alone).



Live web app Grad-CAM overlay identifying critical facial region attention.

[CORRECT] kaggle_input_datasets_ahnuf05_ai-image-with-nano-banana-2-0-vs-real-image_dataset_test_fake_fake_00151.jpg | Prediction: Fake | True: Fake
Real: 14.97% | Fake: 85.03%



Parallel Forensic Meta-Detector Module

Non-CNN Signals

To complement CNN deep learning predictions, our backend runs a parallel non-neural forensic meta-detector (meta_detector.py) scanning raw image bytes.

EXIF Metadata

Camera make, software history, GPS, and C2PA markers.

Watermark Scan

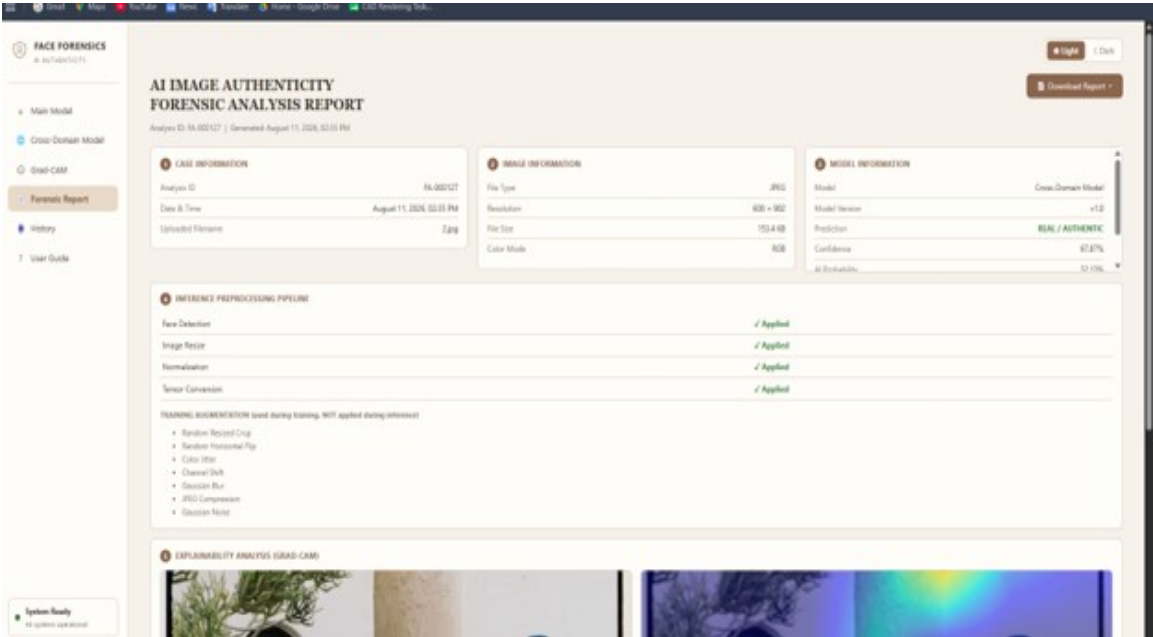
Stable Diffusion, DALL-E 3, and SDXL invisible patterns.

Spectral Analysis

High-frequency FFT energy and grid artifact detection.

ELA Forensics

Error Level Analysis highlights unequal JPEG compression.



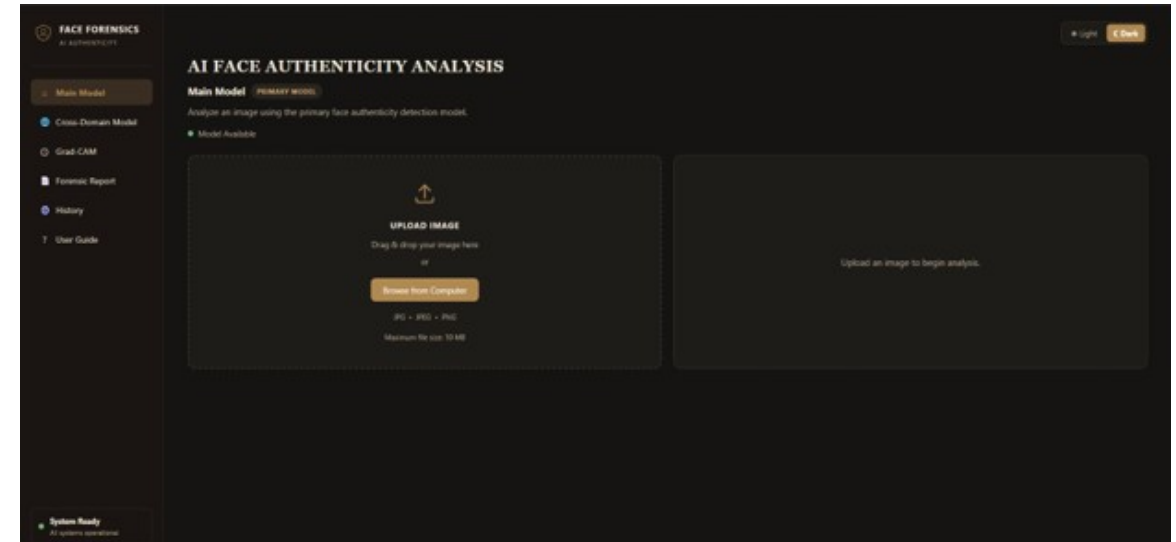
Detailed forensic report layout displaying EXIF tags and ELA results.
Provides independent forensic verification even if deep learning weights fail due to domain shift.

Cloud Architecture & Web Application

LIVE SYSTEM

The web application is fully containerized and hosted on Render Cloud Infrastructure, serving real-time requests globally.

- ✓ **Backend Framework:** FastAPI + Uvicorn with asynchronous endpoints (/predict, /report, /report/docx, /health).
- ✓ **Containerization:** Dockerized build utilizing lightweight Linux base image to fit 512 MB RAM limits.
- ✓ **Application Pages:** 5 complete web pages — Main Analysis, Cross-Domain Mode, Grad-CAM Viewer, Forensic Report Generator, and History Tracker.
- ✓ **Live URL:** <https://group-11-ds-and-ai-lab-project.onrender.com>

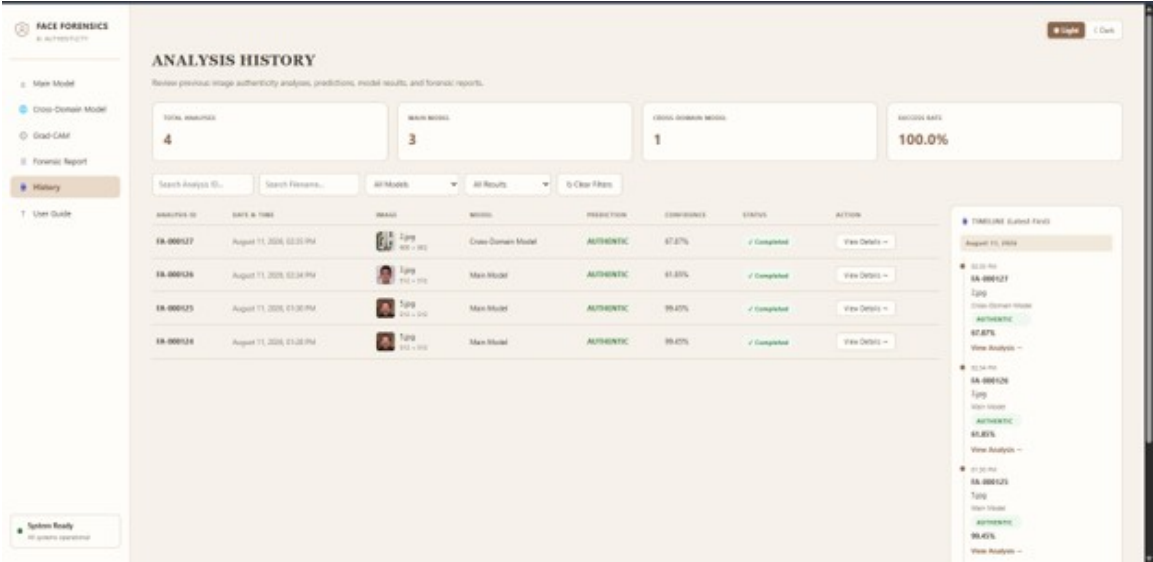


Live web app dashboard showing authenticity verdict and confidence meter.

Milestone 6 Final Deliverables Checklist

Project Completion

- ✓ Final Slide Presentation Deck (This Deck)
- ✓ Comprehensive Technical Report (Docx & PDF)
- ✓ Non-Technical Executive Summary
- ✓ Complete User & Developer Manuals
- ✓ Live Cloud Deployment on Render
- ✓ Team Contribution Summary Matrix



Prediction History module tracking previous user requests and verdicts.

Key Engineering Takeaways & Limitations

FINAL ASSESSMENT

👍 Key Project Strengths

- ✓ **High In-Domain Accuracy:** 95.98% deployed acc / 99.63% research model acc.
- ✓ **Zero Missed Fakes:** 100% Fake Recall on test dataset (Security First).
- ✓ **OOD Generalization:** 80% on ChatGPT/Gemini faces - Raunak's M3 track (per team report; eval script not preserved).
- ✓ **Multi-Signal Forensics:** Integrated Grad-CAM + EXIF/Watermark metadata.
- ✓ **Live Production:** Responsive web application deployed on Docker cloud.

⚠️ Honest Limitations

- ⚠️ **Smartphone Domain Shift:** 8.6% accuracy on Real-Latest probe due to ISP HDR shortcut learning.
- ⚠️ **Augmentation Gap:** ColorJitter failed to simulate non-linear camera sharpening.
- ⚠️ **Latency Trade-Off:** Grad-CAM computation adds ~2.2s execution overhead per request.
- ⚠️ **Hardware Limits:** Free-tier 512MB RAM caps hostable models to 2.
- ⚠️ **Demographic Bias:** Fairness across demographic groups was never audited (Medium severity per M5/M6 reports).

Core Lesson: High benchmark accuracy does not equal real-world reliability. Honest error reporting is essential for deployed AI safety.

FUTURE WORK ROADMAP

Thank You! Questions & Discussion

1. HDR Data Augmentation

Develop specialized tone-mapping ISP transformations in PyTorch training loop.

2. Hard-Negative Mining

Collect 10k modern smartphone photos to eliminate shortcut bias.

3. Video Frame Forensics

Extend pipeline to temporal frame-by-frame deepfake video analysis.

Live Application: <https://group-11-ds-and-ai-lab-project.onrender.com>

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